

Zachariah Newbery

Final-year computer science student at the University of St. Andrews with a natural curiosity nurtured through self-motivated learning with like-minded students to learn multiple languages, alongside a broad academic and linguistic foundation.

Contact

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[LinkedIn Profile](#)

EDUCATION

University of St. Andrews, St. Andrews — MSci (Hons) Computer Science

GRADUATION DATE: JUN 2023

5th Year: Masters Programming Project, Machine Learning, Software Architecture, Parallelism and Concurrency, Interactive Hardware/Software

4th Year: Programming Language Design, Security, Signal Processing, Software Programming Principles, Data Visualisation, Natural Language Processing, Computer Architecture, Video Game Design

Weighted Average: 18.2/20 (High First Class)

SJI International High School, Singapore — International Baccalaureate

NOV 2017

Grade: 43/45 (Diploma Awarded). Higher Levels: Physics (7), Chemistry (7), Mathematics (7)

EXPERIENCE

Google, Dublin — Site Reliability Engineering Intern

MAY 2022 - AUG 2022

- Developed a HTTP web server in **Go** serving HTML templates integrated with a **TypeScript** frontend framework to facilitate resource planning for ML models along variable axes of resource management, reducing the time spent on resource planning by weeks per quarter, improving cost accuracy and thereby resource utilisation.
- Presented to a director and multiple stakeholders to improve adoption and buy-in of the tool, alongside running brainstorming sessions to facilitate design decisions and feature improvements from a stakeholders perspective.

ARM, Cambridge — PTUG Software Engineering Intern (GPU)

MAR 2021 - MAY 2022

- Professional skills development undertaken part-time during academic semesters through specialist talks and recommended courses.
- Added features using **Python** and **Typescript** to offline GPU tooling and web interfaces to work with flow control and post-processing analysis of GPU during the summer.

University of St. Andrews, St. Andrews — Undergraduate Research Assistant

SEPT 2020 - MAY 2022

- Implemented data visualisation within a biological evolutionary simulation (S.E.E) using **C#** within Unity, alongside a GUI for user-guided customisation.
- Improved **Docker** containerisation for a Jupyter kernel for GAP, a mathematical programming language.

PROJECTS

CS4303 Project 4 — Wanderer's Path

Developed a 2D novel top-down RPG video game in **Java** as an individual project to develop skills related to game design, user experience, procedural level generation, searching algorithms, ray-casting and behavioural AI algorithms.

CS3099 — Federated Social Media (Backend)

Developed in **Rust** as part of a 5-person group to develop skills related to agile software development, federation and web hosting. Implemented functionality includes user accounts and authentication, communities, posts, searching and federated lookup.

SKILLS

- Written/Verbal Communication
- Teamwork
- Critical Thinking
- Problem Solving
- Leadership

AWARDS

University of St. Andrews

Dean's List (1st, 2nd, 3rd Year):

Attained a first-class mean module grade.

Christopher Strachey Prize: Awarded to the best student in CS2001 (Algorithms and Data Structures).

Supplementary Medal: Awarded for performance in CS2001.

Professional Skills Curriculum Award.

University of St. Andrews Athletic Union

Half Blue: Awarded for activities in Taekwondo.

Taekwondo

3rd Degree Black Belt.

2nd Degree Kukkiwon.

Bronze and Gold Medals at National Competitions.

ABRSM

Grade 5 Saxophone.

National Youth Achievement Award Gold (DoE award equivalent).

LANGUAGES

Spoken: English, Spanish, Mandarin Chinese.

Programming: Java, C, Go, Python, Haskell, SQL, Rust, TypeScript, C#.

Markup: HTML, CSS, LaTeX, AST.

Systems: Windows, Linux.